HI $\times$ UGRB Cross-Correlation Pipeline

Complete Physics Reference

Research Progress Update

April 20, 2026

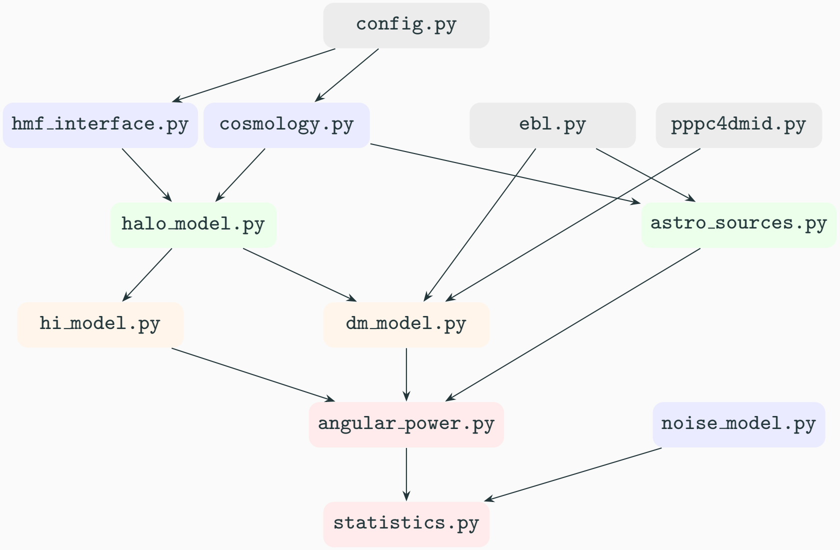
Outline

Pipeline Overview

Scientific Goal

- Cross-correlate **HI 21-cm intensity maps** (MeerKAT/SKA) with the **unresolved gamma-ray background** (Fermi-LAT)
- Detect astrophysical cross-signal (blazars, mAGN, SFGs sharing LSS with HI)
- Constrain dark matter annihilation via excess $C_\ell^{\mathrm{HI}\times\gamma}$
- Based on Pinetti, Camera, Fornengo & Regis (2020, JCAP 07:044)

Module Dependency DAG



Unit Conventions

Quantity	Unit	Convention
Distances	Mpc/h (comoving)	χ, R_{vir}, r_s
Masses	$M_\odot/h$	M, M_{HI}
Wavenumbers	h/Mpc	k
Power spectra	$(\mathrm{Mpc}/h)^3$	P_{lin}, P_{HI}
Mass function	$(\mathrm{Mpc}/h)^{-3}\,(M_\odot/h)^{-1}$	dn/dM
Brightness temp.	mK	$\bar{T}_b$
Luminosities	erg/s	GLFs
Fermi noise	$\mathrm{cm}^{-4}\,\mathrm{s}^{-2}\,\mathrm{sr}^{-1}$	N^γ

All internal calculations use h-dependent comoving units.

Physical units appear only at module boundaries (explicit conversion).

Cosmology

Planck 2018 Parameters

$$H_0 = 67.36\ \mathrm{km/s/Mpc}$$
$$h = 0.6736$$
$$\Omega_\mathrm{B} h^2 = 0.02237$$
$$\Omega_c h^2 = 0.1200$$
$$n_s = 0.9649$$

$$\sigma_8 = 0.8111$$
$$A_s \approx 2.1 \times 10^{-9}$$
$$\Omega_\mathrm{M} = 0.3153$$
$$\tau = 0.0544$$
$$T_\mathrm{CMB} = 2.7255\ \mathrm{K}$$

TT,TE,EE+lowE+lensing best-fit (Planck Collaboration 2020, A&A 641, A6).

Hubble Rate and Distances

$$E(z) = \sqrt{\Omega_{\mathrm{M}}(1+z)^3 + \Omega_\Lambda} \tag{1}$$

$$H(z) = H_0\, E(z) \tag{2}$$

$$\chi(z) = \frac{c}{H_0}\int_0^z \frac{dz'}{E(z')} \quad [\,\mathrm{Mpc}/h] \tag{3}$$

$$d_L(z) = (1+z)\,\chi(z) \tag{4}$$

Implementation: $\chi(z)$ uses a cumulative-Simpson spline (4096 pts, $\sim 10^{-13}$ accuracy). Mutation guard re-initialises CAMB if cosmology parameters change.

Linear Power Spectrum and Growth

$$P_{\mathrm{lin}}(k, z) \quad \text{via CAMB Boltzmann solver} \tag{5}$$

$$D(z) \propto E(z) \int_z^\infty \frac{(1+z')\,dz'}{E(z')^3}, \quad D(0) = 1 \tag{6}$$

$$\sigma^2(R, z) = \frac{1}{2\pi^2} \int_0^\infty dk\, k^2\, P_{\mathrm{lin}}(k, z)\, W^2(kR) \tag{7}$$

$$W(x) = \frac{3(\sin x - x\cos x)}{x^3} \quad \text{(top-hat)} \tag{8}$$

Key identity: $R(M) = \left(\frac{3M}{4\pi\bar{\rho}_m}\right)^{1/3}$ links mass to smoothing scale.

Halo Model

Virial Overdensity and Halo Structure

Bryan & Norman (1998):

$$\Delta_{\mathrm{vir}}(z) = 18\pi^2 + 82x - 39x^2, \quad x = \Omega_m(z) - 1 \tag{9}$$

Virial radius and circular velocity:

$$R_{\mathrm{vir}} = \left(\frac{3M}{4\pi\, \Delta_{\mathrm{vir}}(z)\, \rho_{\mathrm{c}}(z)} \right)^{1/3} \quad [\,\mathrm{Mpc}/h] \tag{10}$$

$$v_c = \sqrt{\frac{GM_{\mathrm{phys}}}{R_{\mathrm{phys}}}} \quad [\mathrm{km/s}] \tag{11}$$

Physical units: $M_{\mathrm{phys}} = M/h$, $R_{\mathrm{phys}} = R_{\mathrm{vir}}/h$.

Sheth–Mo–Tormen Mass Function and Bias

Peak height: $\nu = \delta_\mathrm{c}^2/\sigma^2(M,z), \quad \delta_\mathrm{c} = 1.686$

Mass function (SMT 2001):

$$\nu f(\nu) = A\left[1 + (q\nu)^{-p}\right]\sqrt{\frac{q\nu}{2\pi}}\,\exp\left(-\frac{q\nu}{2}\right) \tag{12}$$

$q = 0.707,\ p = 0.3,\ A \approx 0.3222$

Halo bias (Sheth & Tormen 1999):

$$b(\nu) = 1 + \frac{q\nu - 1}{\delta_\mathrm{c}} + \frac{2p}{\delta_\mathrm{c}\left[1 + (q\nu)^p\right]} \tag{13}$$

NFW Profile and Concentration

Correa et al. (2015) concentration–mass:

$$\log_{10} c_{200} = \alpha + \beta \log_{10}\left(\frac{M}{M_\odot}\right)\left[1 + \gamma\left(\log_{10}\frac{M}{M_\odot}\right)^2\right] \tag{14}$$

with z-dependent (α, β, γ) polynomials (Planck Appendix B1).
Converted $c_{200} \to c_{\mathrm{vir}}$ via Newton iteration on the implicit NFW relation.

Analytic NFW Fourier transform:

$$\tilde{u}(k|M) = \frac{1}{f(c)}\left[\sin(kr_s)\left(\mathrm{Si}_{1+c} - \mathrm{Si}_1\right) + \cos(kr_s)\left(\mathrm{Ci}_{1+c} - \mathrm{Ci}_1\right) - \frac{\sin(ckr_s)}{(1{+}c)kr_s}\right] \tag{15}$$

where $f(c) = \ln(1 + c) - c/(1 + c)$ and $r_s = R_{\mathrm{vir}}/c$.

HI Model

HI Mass–Halo Mass Relation

Padmanabhan, Refregier & Amara (2017), modified NFW fit:

$$M_{\mathrm{HI}}(M,z) = \alpha\, f_{H,c}\, M \left(\frac{M}{10^{11} h^{-1} M_\odot}\right)^{\beta} \exp\left[-\left(\frac{v_{c,0}}{v_c(M,z)}\right)^{3}\right] \tag{16}$$

α	β	$v_{c,0}$ [km/s]	$c_{\mathrm{HI},0}$	γ_c
0.176	−0.69	40.7	139	0.13

$f_{H,c} = (1 - Y_P)\Omega_{\mathrm{B}}/\Omega_{\mathrm{M}}, \quad Y_P = 0.24$

HI concentration: $c_{\mathrm{HI}} = c_{\mathrm{HI},0}\,(M/10^{11} M_\odot)^{-0.109} \cdot 4/(1+z)^{\gamma_c}$

Note: mass pivot is $10^{11} M_\odot$ (physical), not $M_\odot/h$.

Modified NFW HI Profile

$$\rho_{\rm HI}(r) = \frac{\rho_0 \, r_s^3}{(r + 0.75\, r_s)(r + r_s)^2} \tag{17}$$

Normalized: $\int_0^{R_{\rm vir}} 4\pi r^2 \, \rho_{\rm HI} \, dr = M_{\rm HI}$

Analytic normalization via partial fractions (A=9, B=−8, C=−4):

$$\int_0^c \frac{x^2 \, dx}{(x+0.75)(x+1)^2} = 9 \ln \frac{c+0.75}{0.75} - 8 \ln(c+1) + 4\left(\frac{1}{c+1} - 1\right) \tag{18}$$

Fourier transform: Numerical quadrature

$$\tilde{u}_{\rm HI}(k|M) = \frac{4\pi}{M_{\rm HI}} \int_0^{R_{\rm vir}} r^2 \, \rho_{\rm HI}(r) \, \frac{\sin kr}{kr} \, dr \tag{19}$$

Brightness Temperature and HI Window

Mean comoving HI density and bias:

$$\bar{\rho}_{\mathrm{HI}}(z) = \int \frac{dn}{dM} \, M_{\mathrm{HI}}(M, z) \, dM \tag{20}$$

$$\Omega_{\mathrm{HI}}(z) = \bar{\rho}_{\mathrm{HI}} / \rho_{\mathrm{c}} \quad \text{(D5: computed, not fixed)} \tag{21}$$

$$b_{\mathrm{HI}}(z) = \frac{1}{\bar{\rho}_{\mathrm{HI}}} \int \frac{dn}{dM} \, M_{\mathrm{HI}} \, b(M, z) \, dM \tag{22}$$

Brightness temperature:

$$\bar{T}_b(z) = 188 \, h \, \Omega_{\mathrm{HI}}(z) \, \frac{(1+z)^2}{E(z)} \, \mathrm{mK} \tag{23}$$

Window function (per-χ):

$$W_{\mathrm{HI}}(\chi) = \bar{T}_b(z) \, \phi(z) \, \frac{H(z)}{c \, h}, \quad \phi(z) = \frac{1}{z_{\mathsf{max}} - z_{\mathsf{min}}} \tag{24}$$

Bias enters through P_{HI}, **not** through W_{HI}.

Cunnington Brightness Mode (D15)

Alternative for MeerKLASS data analyses (Cunnington+2025):

$$\Omega_{\mathrm{HI}}^{\mathrm{C}}(z) = 6.7432\times10^{-4} + 3.9\times10^{-4}\,z - 6.5\times10^{-5}\,z^2 \tag{25}$$

$$b_{\mathrm{HI}}^{\mathrm{C}}(z) = 0.842 + 0.693\,z - 0.0459\,z^2 \tag{26}$$

$$\bar{T}_b^{\mathrm{C}}(z) = 180\,h\,\Omega_{\mathrm{HI}}^{\mathrm{C}}(z)\,\frac{(1+z)^2}{E(z)}\,\mathrm{mK} \tag{27}$$

Key differences from Padmanabhan mode:

- 180 mK prefactor (Battye+2013) vs 188 mK
- Polynomial Ω_{HI}, b_{HI} vs halo-model integrals
- $P_{\mathrm{HI}}^{2h} \to b_{\mathrm{HI}}^2\,P_{\mathrm{lin}}$ (no halo integral)

Dispatched via per-telescope `default_hi_brightness` field.

Dark Matter Model

NFW ρ^2 Profile and Clumping Factor

Scale density and ρ^2 integral:

$$\rho_s = \frac{M}{4\pi\, r_s^3\, f(c)} \tag{28}$$

$$\int_0^{R_{\rm vir}} 4\pi r^2\, \rho_{\rm NFW}^2\, dr = \frac{4\pi}{3}\, \rho_s^2\, r_s^3 \left[1 - \frac{1}{(1+c)^3}\right] \tag{29}$$

Moliné et al. (2017) substructure boost:

$$\log_{10} B(M, z{=}0) = \sum_{i=0}^{5} b_i \left[\log_{10} \frac{M}{M_\odot}\right]^i, \quad B(M,z) = \frac{B(M,0)}{1+z} \tag{30}$$

Clumping factor (physical-frame definition):

$$\Delta^2(z) = \frac{1}{\bar{\rho}_m^2\,(1+z)^3} \int \frac{dn}{dM} \left[1 + B(M,z)\right] \int \rho^2\, d^3x\, dM \tag{31}$$

$(1+z)^3$ Bookkeeping

Why the $(1+z)^3$ factors?

1. **Clumping:** Halo integral gives comoving $\langle\rho^2\rangle_{\rm com}$.
   Divide by $\bar{\rho}^2_{\rm com}\cdot(1+z)^3$ to get physical $\Delta^2_{\rm phys}$.
2. **Window function:** Annihilation rate $\propto \rho^2_{\rm phys} = \rho^2_{\rm com}\cdot(1+z)^6$.
   Per comoving volume: divide by $(1+z)^3$.
   Net factor in $W_\gamma^{\rm DM}$: $(1+z)^3$.
3. **End-to-end:** $\frac{\Delta^2}{(1+z)^3}\times(1+z)^3$ in $W_\gamma^{\rm DM}$ recovers the comoving integral.

The factors do **not** simply cancel — they serve different physical roles.

DM Annihilation Window Function

$$W_\gamma^{\mathrm{DM}}(\chi) = \frac{\langle \sigma v \rangle}{8\pi} \left(\frac{\rho_{\mathrm{DM}}}{m_\chi} \right)^2 (1+z)^3 \, \Delta^2(z) \, \frac{dN}{dE'} \, e^{-\tau(E,z)} \tag{32}$$

- $E' = (1+z)\, E_{\mathrm{obs}}$ (rest-frame for PPPC spectrum)
- $\tau(E_{\mathrm{obs}}, z)$ (EBL at **observed** energy)
- No baked-in $1/H(z)$ — Limber measure supplies $d\chi/dz$
- $\rho_{\mathrm{DM}} = \Omega_{\mathrm{DM}} \cdot \rho_{\mathrm{c}}$ (comoving, converted to GeV/cm^3)

Units: [photons $(\mathrm{Mpc}/h)^{-3}\ \mathrm{s}^{-1}\ \mathrm{sr}^{-1}\ \mathrm{GeV}^{-1}$]

Astrophysical Sources

Four Source Populations

Source	GLF method	α (photon index)	Reference
BL Lac	LDDE inverse-sum	2.11	Ajello+2014
FSRQ	LDDE inverse-sum	2.44	Ajello+2012
mAGN	Radio$\rightarrow$gamma chain	2.37	Di Mauro+2014
SFG	IR$\rightarrow$gamma chain	2.70	Gruppioni+2013

All share the same window function structure $W_\gamma^{\rm astro}$, differing only in their GLF $\Phi(L, z)$.

LDDE Luminosity Function (Blazars)

Double power-law:

$$\frac{d\Phi}{d\log_{10} L} = \frac{A}{(L/L_c)^{\gamma_1} + (L/L_c)^{\gamma_2}} \tag{33}$$

Smooth inverse-sum evolution (Ajello+2012 Eq. 15, D12):

$$e(z, L) = \left[r^{-p_1} + r^{-p_2}\right]^{-1}, \quad r = \frac{1+z}{1+z_c(L)} \tag{34}$$

with $z_c(L) = z_c^* \, (L/L_{\mathrm{ref}})^{\alpha}$.

	A	L_c [erg/s]	γ_1	γ_2	z_c^*	p_1	p_2
FSRQ	3.06×10^{-9}	8.4×10^{47}	0.21	1.58	1.47	7.35	-6.51
BL Lac	9.20×10^{-11}	2.43×10^{48}	1.12	3.71	1.67	4.50	-12.88

mAGN: Radio→Gamma Conversion Chain

4-step chain (Di Mauro+2014 / Pinetti 2022 Eq. C.19):

1. $L_\gamma \to \nu L_{\nu,\mathrm{core}}^{5\mathrm{GHz}}$: $\log L_\gamma = 2.0 + 1.008 \log(\nu L_\nu)$ [erg/s]
2. $\nu L_\nu \to L_{\mathrm{core}}^{5\mathrm{GHz}}$: divide by $\nu \cdot 10^7$ [W/Hz]
3. $L_{\mathrm{core}} \to L_{\mathrm{tot}}^{1.4\mathrm{GHz}}$: Lara (2004): $\log L_{\mathrm{core}} = 4.2 + 0.77 \log L_{\mathrm{tot}}$
4. $L_{\mathrm{tot}}^{1.4} \to L_{151}$: $(1400/151)^{-0.80}$ [Inoue 2011]

Assembled GLF:

$$\phi_\gamma = \frac{k\,\eta(z)}{(1+z)^{2-\Gamma}} \frac{\rho_r(L_{151}, z)}{\ln 10 \cdot L_{151}} \left| \frac{dL_{151}}{dL_\gamma} \right| \tag{35}$$

$k = 3.05$, $\Gamma = 2.37$. $\eta(z)$ = Willott→Planck volume correction.

SFG: IR→Gamma Chain

Gruppioni+2013 modified Schechter (3 components):

$$\phi_i = \phi_{0,i}(z) \left(\frac{L_{\rm IR}}{L_{0,i}}\right)^{1-\gamma_i} \exp\left[-\frac{\log_{10}^2(1 + L_{\rm IR}/L_{0,i})}{2\sigma_i^2}\right] \tag{36}$$

$\phi_{\rm IR} = \phi_{\rm spiral} + \phi_{\rm starburst} + \phi_{\rm SF-AGN}$

Ackermann+2012 L_γ–$L_{\rm IR}$ scaling:

$$\log_{10} L_\gamma = 1.09 \log_{10}\left(\frac{L_{\rm IR}}{10^{10} L_\odot}\right) + 39.19 \tag{37}$$

SFG GLF:

$$\phi_\gamma^{\rm SFG} = \phi_{\rm IR} \cdot \frac{|d \log L_{\rm IR}/d \log L_\gamma|}{L_\gamma \ln 10} \tag{38}$$

Astrophysical Window Function

$$W_\gamma^{\rm astro}(\chi) = \frac{1}{4\pi h^3} \int_{L_{\rm min}}^{L_{\rm up}} \Phi(L,z)\, \frac{L}{E_{\rm GeV\to erg}\, I_\alpha}\, E_{\rm rest}^{-\alpha}\, dL \;\times e^{-\tau(E,z)} \tag{39}$$

- $E_{\rm rest} = (1+z)\,E_{\rm obs}$ (K-correction inside window)
- $1/h^3$ converts Mpc^{-3} to (Mpc$/h)^{-3}$
- $L_{\rm up} = \min(L_{\rm max},\, L_{\rm sens}(z))$ for unresolved sources
- D14: EBL attenuation $e^{-\tau}$ at observed energy

$F_{\rm sens}$ dispatch (D17):

- `'pinetti_constant'`: $F_{\rm sens} = 10^{-10}\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$ (Pinetti+2020)
- `'4fgl_dr4_psf'`: $F_{\rm sens} = 7.3{\times}10^{-11}\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$ (Ballet+2023) + PSF scaling

EBL and PPPC4DMID

Extragalactic Background Light Opacity

Dominguez et al. (2011) tabulated opacity via `ebltable`:

$$\tau(E_{\rm obs}, z) \quad \text{(pair-production optical depth)} \tag{40}$$

$$A(E_{\rm obs}, z) = e^{-\tau(E_{\rm obs}, z)} \quad \text{(attenuation factor)} \tag{41}$$

Key convention: E is the **observed** photon energy (at Earth), not rest-frame.

Available models: `dominguez` (default), `franceschini`, `finke`, `saldana-lopez21`.

Where EBL enters:

- $W_\gamma^{\rm DM}$: attenuates DM annihilation photons
- $W_\gamma^{\rm astro}$: attenuates astrophysical source photons (D14)
- $L_{\rm sens}$: raises detection threshold at high z via $J_\alpha^{\rm EBL}(z)$

PPPC4DMID Photon Yields

Cirelli et al. (2011): Tabulated $dN/d\log_{10} x$ per annihilation event.

$$x = E/m_\chi \tag{42}$$

$$\frac{dN}{dx} = \frac{dN/d\log_{10} x}{x \ln 10} \tag{43}$$

$$\frac{dN}{dE} = \frac{1}{m_\chi}\frac{dN}{dx} \tag{44}$$

Channels used: $b\bar{b}$ (primary), $\tau^+\tau^-$, W^+W^-, ZZ, $c\bar{c}$, $t\bar{t}$, gg, hh, etc.

Mass range: 5 GeV – 100 TeV.

D7: Pipeline uses public PPPC4DMID tables (thesis used private Pythia code).

Noise and Beams

Radio System Temperature

Pinetti generic (Camera+2013; legacy MeerKAT/SKA):

$$T_{\rm sys}^{\rm P}(\nu) = 30\ \mathrm{K} + 60\ \mathrm{K}\left(\frac{300\ \mathrm{MHz}}{\nu}\right)^{2.55} \tag{45}$$

MeerKAT canonical (Cunnington+2025 Eq. A19; D16):

$$T_{\rm sys}^{\rm M}(\nu) = T_{\rm rx}(\nu) + T_{\rm spl} + T_{\rm CMB} + T_{\rm gal}(\nu) \tag{46}$$

$$\begin{aligned} T_{\rm rx} &= 7.5 + 10\,(\nu/\mathrm{GHz} - 0.75)^2\ \mathrm{K} \\ T_{\rm spl} &= 3\ \mathrm{K}, \quad T_{\rm CMB} = 2.725\ \mathrm{K} \\ T_{\rm gal} &= 15\ \mathrm{K}\,(408\ \mathrm{MHz}/\nu)^{2.75} \end{aligned}$$

Gives ∼14 K (L-band), ∼16 K (UHF) — half the Pinetti formula.
Dispatched via `T_sys_model`: `'pinetti'` or `'meerkat'`.

Radio Noise Power Spectra

Single-dish:

$$N_{\mathrm{dish}} = \frac{T_{\mathrm{sys}}^2\,\Omega_S}{N_d\,t\,\Delta\nu\,N_b\,N_{\mathrm{pol}}\,\eta^2} \quad [\mathrm{mK}^2\ \mathrm{sr}] \tag{47}$$

Interferometer:

$$N_{\mathrm{interf}}(\ell) = \frac{T_{\mathrm{sys}}^2\,\Omega_S\,\mathrm{FoV}}{n(u)\,t\,\Delta\nu\,N_b\,N_{\mathrm{pol}}\,\eta^2} \tag{48}$$

Valid for $\ell \geq \ell_{\mathrm{cut}} = \pi D_{\mathrm{short}}/(1.22\,\lambda)$

Combined: $N_\ell^{\mathrm{radio}} = \min(N_{\mathrm{dish}},\, N_{\mathrm{interf}})$ for $\ell \geq \ell_{\mathrm{cut}}$

Beam: $B_\ell^{\mathrm{HI}} = \exp(-\ell^2\sigma_{\mathrm{beam}}^2/2), \quad \sigma_{\mathrm{beam}} = 1.22\lambda/(D\sqrt{8\ln 2})$

Fermi-LAT PSF and Beam

Gaussian approximation (forecast mode, Pinetti+2020):

$$\sigma_0(E) = 1.20\,(E/0.5\,\mathrm{GeV})^{-0.95} + 0.05 \tag{49}$$

$$\sigma_b(\ell) = \sigma_0/(1 + 0.25\,\sigma_0\,\ell) \tag{50}$$

$$B_\ell^\gamma = \exp(-\sigma_b^2\,\ell^2/2) \tag{51}$$

Exact King PSF (data mode, Ammazzalorso+2018):

$$\mathrm{PSF}(\theta) = \frac{1 - 1/\gamma}{2\pi\sigma_K^2}\left[1 + \frac{\theta^2}{2\gamma\sigma_K^2}\right]^{-\gamma}, \quad \gamma = 2 \tag{52}$$

$$W_\ell(E) = 2\pi \int_0^\pi \mathrm{PSF}(\theta)\, P_\ell(\cos\theta)\, \sin\theta\, d\theta \tag{53}$$

Energy-averaged: $\langle W_\ell^k \rangle = \int W_\ell(E)\, E^{-\alpha}\, dE \;/\; \int E^{-\alpha}\, dE, \quad \alpha = 2.3$

Limber Integration

Cross-Power Spectra (Halo Model)

HI $\times$ DM 2-halo (Pinetti Eq. 5.2):

$$P^{2h}_{\mathrm{HI}\times\mathrm{DM}}(k,z) = \underbrace{\left[\int \frac{dn}{dM}\, b\, \frac{\tilde{v}}{\Delta^2}\, dM\right]}_{\text{DM kernel}} \underbrace{\left[\int \frac{dn}{dM}\, b\, \frac{\tilde{u}_{\mathrm{HI}}\, M_{\mathrm{HI}}}{\bar{\rho}_{\mathrm{HI}}}\, dM\right]}_{\text{HI kernel}} P_{\mathrm{lin}}(k,z) \tag{54}$$

HI $\times$ Astro 2-halo (Pinetti Eq. 5.4):

$$P^{2h}_{\mathrm{HI}\times\mathrm{astro}}(k,z) = \left[\int \frac{dn}{dM}\, b\, \frac{\tilde{u}_{\mathrm{HI}}\, M_{\mathrm{HI}}}{\bar{\rho}_{\mathrm{HI}}}\, dM\right] b_{\mathrm{astro}}\, P_{\mathrm{lin}}(k,z) \tag{55}$$

Cunnington mode: HI kernel $\rightarrow b^{\mathrm{C}}_{\mathrm{HI}}(z)$ (polynomial), no mass integral.

Limber Integral

$$\boxed{C_\ell^{ij} = \int \frac{d\chi}{\chi^2}\, W_i(\chi)\, W_j(\chi)\, P_{ij}\left(k = \frac{\ell + \frac{1}{2}}{\chi},\ z\right)} \tag{56}$$

D8: $k = (\ell + \frac{1}{2})/\chi$ (LoVerde & Afshordi 2008) vs thesis $k = \ell/\chi$.

Implementation:

- Riemann sum over redshift grid ($n_z = 200$ points in $[z_{\mathrm{min}}, z_{\mathrm{max}}]$)
- Weight: $(d\chi/dz)/\chi^2 \cdot \Delta z$ where $d\chi/dz = c\,h/H(z)$ $[\mathrm{Mpc}/h]$
- Product: $W_{\mathrm{HI}} \times W_\gamma \times P_{\mathrm{cross}}(k, z) \times$ weight

Multi-energy optimisation: The kernel $(\chi, H, W_{\mathrm{HI}}, P_{\mathrm{cross}})$ is E-independent. Only $W_\gamma(E, z)$ varies with energy. Compute kernel once, loop over E in inner loop.

Dimensional Analysis of C_ℓ

Factor	Units
$d\chi/dz$	$[\mathrm{Mpc}/h]$
$1/\chi^2$	$[(\mathrm{Mpc}/h)^{-2}]$
W_{HI}	$[\mathrm{mK}\cdot(\mathrm{Mpc}/h)^{-1}]$
W_γ	$[\mathrm{ph}\ (\mathrm{Mpc}/h)^{-3}\ \mathrm{s}^{-1}\ \mathrm{sr}^{-1}\ \mathrm{GeV}^{-1}]$
P_{cross}	$[(\mathrm{Mpc}/h)^3]$
dz	$[1]$
$C_\ell^{\mathrm{HI}\times\gamma}$	$[\mathrm{mK}\cdot\mathrm{ph}\cdot(\mathrm{Mpc}/h)^{-2}\cdot\mathrm{s}^{-1}\cdot\mathrm{sr}^{-1}\cdot\mathrm{GeV}^{-1}]$

The $(\mathrm{Mpc}/h)^{-2}$ is the angular projection factor.
Converting to cm gives the standard cross-power units.

Statistics

Gaussian Variance

Noise-dominated approximation (Pinetti Eq. 5.5):

$$(\Delta C_\ell)^2 = \frac{1}{(2\ell+1)\, f_{\mathrm{sky}}} \frac{N^\gamma}{(B_\ell^\gamma)^2} \left[C_\ell^{\mathrm{HI,HI}} + \frac{N^{\mathrm{HI}}}{(B_\ell^{\mathrm{HI}})^2} \right] \tag{57}$$

Unit conversion: $N^\gamma\ [\mathrm{cm}^{-4}\,\mathrm{s}^{-2}\,\mathrm{sr}^{-1}] \to [(\,\mathrm{Mpc}/h)^{-4}\,\mathrm{s}^{-2}\,\mathrm{sr}^{-1}]$ via $\times\,(L^{\mathrm{cm}}_{\mathrm{Mpc}/h})^4$

Effective sky fraction: $f_{\mathrm{sky}} = \min(f_{\mathrm{sky}}^{\mathrm{radio}},\, f_{\mathrm{sky}}^{\gamma})$

Signal-to-Noise Ratio

$$\mathrm{SNR}^2 = \sum_{\ell,\, E\text{-bins}} \left[\frac{C_\ell^{\mathrm{HI}\times\gamma_\star}}{\Delta C_\ell} \right]^2 \tag{58}$$

- Numerator: astrophysical-only signal (sum of 4 source classes)
- Denominator: noise-dominated variance
- Sum over 12 Fermi energy bins (forecast) or 11 Ammazzalorso bins (data)
- ℓ-range: $[10, 1000]$ (forecast) or per-bin $[\ell_{\mathsf{min}}, \ell_{\mathsf{max}}(E)]$ (data)

Per-telescope dispatch: `compute_SNR` reads `default_hi_brightness` and `default_unresolved_mode` from the telescope config entry.

DM Exclusion Curves

$\Delta\chi^2$ **test statistic** (Pinetti Eq. 5.7):

$$\Delta\chi^2 = \sum_{\ell,E} \left[\frac{C_\ell^{\mathrm{astro+DM}}}{\Delta C_\ell}\right]^2 - \left[\frac{C_\ell^{\mathrm{astro}}}{\Delta C_\ell}\right]^2 \tag{59}$$

Exclusion scaling: Since $C_\ell^{\mathrm{DM}} \propto \langle\sigma v\rangle$:

$$\langle\sigma v\rangle^{\mathrm{excl}} = \langle\sigma v\rangle^{\mathrm{test}} \sqrt{\frac{\Delta\chi^2_{\mathrm{threshold}}}{\Delta\chi^2_{\mathrm{test}}}} \tag{60}$$

- 95% CL: $\Delta\chi^2_{\mathrm{thr}} = 4.0$
- 5σ: $\Delta\chi^2_{\mathrm{thr}} = 25.0$

Scan over $m_\chi \in [10\,\mathrm{GeV},\ 10\ \mathrm{TeV}]$ to produce the exclusion curve.

Deviations from Literature

Active Deviations Registry

#	Deviation	Nature
D2	Correa $c(M,z)$ coefficients	Planck App. B1 fit (not thesis cosmology)
D4	SF-AGN k_{R2} sign	Thesis typo corrected (-3.17, not $+3.17$)
D5	Ω_{HI} computed	Halo integral vs fixed 2.45×10^{-4}
D6	SMT $q = 0.707$	Original SMT 1999; thesis uses $q = 0.75$
D7	PPPC4DMID public tables	vs thesis private Pythia code
D8	Limber $k = (\ell+\frac{1}{2})/\chi$	LoVerde & Afshordi 2008 correction
D12	LDDE exponent form	Ajello positive-exponent convention
D14	EBL on astro windows	Now applied; thesis omitted for astro
D15	Cunnington brightness mode	180 mK + polynomial Ω_{HI}, b_{HI}
D16	MeerKAT T_{sys}	15 K Galactic synchrotron (Cunnington+2025)
D17	4FGL-DR4 F_{sens} dispatch	7.3×10^{-11} + PSF scaling (Ballet+2023)

Resolved Deviations

#	Was	Resolution
~~D3~~	No $c_{200} \to c_{\mathrm{vir}}$ conversion	`c200_to_cvir()` now implemented
~~D9~~	$\bar{T}_b$ 188h vs 44 μK form	Mathematically equivalent (convention)
~~D11~~	$\Delta_{\mathrm{vir}} = 200$ fixed	Bryan & Norman z-dependent $\Delta_{\mathrm{vir}}(z)$
~~D13~~	Gruppioni L_0 break uniform	Now spiral-only per Table 8

Cosmology Mismatch Registry

Fit	Paper cosmology	Correction	Residual
Correa $c(M)$	Planck 2013	D2: re-fit	Negligible
Ajello 2012 FSRQ	WMAP-era (h=0.71)	None	$\sim$1–2% in d_L
Ajello 2014 BL Lac	WMAP-era (h=0.71)	None	$\sim$1–2% in d_L
Willott 2001	H_0=50, empty	$\eta(z)$ volume	Shape not re-derived
Gruppioni 2013	H_0=70, Ω_m=0.3	None	Small
Moliné 2017	Planck 2015	None	Weak (N-body)
Padmanabhan 2017	WMAP-3 custom	None	Weak
Cunnington 2025	Planck 2018	None needed	–
Ballet 2023	Cosmology-independent	N/A	–

Willott $\eta(z)$ corrects density only — luminosity rescaling would require re-fitting to original flux data (documented limitation from Di Mauro+2014).